

6. How could the soil be conserved? (B)
[Ans. 2:02:3:02]
7. What is meant by 'Deforestation'? What are the reasons for it? (B) [Ans. 2:03:2 + 2:03:3]
8. Write a short note on each of the following:
 - i) Uses of Forests (C) [Ans. 2:03:1]
 - ii) Afforestation (C) [Ans. 2:03:6]
 - iii) Adverse effects of deforestation (C)
[Ans. 2:03:4]
 - iv) Measures to prevent deforestation (C)
[Ans. 2:03:5]
9. Briefly describe the different types of water resources in India. (B) [Ans. 2:04:1 + 2:04:1:01 to the end of 2:04:1:03]
10. Suggest ways and means of controlling water scarcity. (B) [Ans. 2:04:3]
11. What is meant by 'Mineral Resources'? Briefly explain the different kinds of mineral resources. (B) [Ans. 2:05 + 2:05:1 + 2:05:1:01 to the end of 2:05:1:03]
12. Explain the adverse effects of mining operations. (B) [Ans. 2:05:3]
13. Explain briefly the ways and means of preventing over exploitation of minerals. (B) [Ans. 2:05:4]
14. What are the food resources available in India? (C) [Ans. 2:06]
15. Explain the different food resources and the reasons for decrease in food production. (B)
[Ans. 2:06+ 2:06:1]

16. Briefly explain the 'food crisis' and the factors responsible for it (A) [Ans. 2:06:2]
17. Explain how food production could be enhanced in our country. (B) [Ans. 2:06:2:01]
18. State the uses of energy resources in our life. Discuss briefly the different kinds of energy resources available to us. (A) [Ans. 2:07 + 2:07:2 + 2:07:2:01 + 2:07:2:02]
19. Explain the renewable energy resources. (B)
[Ans. 2:07:2:02]
20. Explain the alternate sources of energy. (B)
[Ans. In 2:07:2:02 I paragraph + 2 to 6 (i.e. except Hydro-electric power)]
21. Write a short note on each of the following :
 - i) Wind energy (C) [Ans. (2) of 2:07:2:02]
 - ii) Solar energy (C) [Ans. (4) of 2:07:2:02]
 - iii) Hydroelectric power (C) [Ans. (1) of 2:07:2:02]
 - iv) Tidal energy (C) [Ans. (5) of 2:07:2:02]
 - v) Biomass energy (C) [Ans. (3) of 2:07:2:02]
 - vi) Geothermal energy (C) [Ans. (6) of 2:07:2:02]
 - vii) Desertification (C) [Ans. 2:02:1:03]
 - viii) Effects of soil erosion (C) [Ans. (C) of 2:02:1:01]
 - ix) Landslides (C) [Ans. 2:02:1:02]
 - x) Measures to prevent desertification (C)
[Ans. 2:02:2]
 - xi) Growing energy needs (B) [Ans. In 2:07:1 paras 1 and 2 only]

Chapter - III

ENVIRONMENTAL POLLUTION, HAZARDS AND DISASTER MANAGEMENT

3:00 Introduction

The basic thesis of growth school is that economic growth is required for political, social and economic stability. Unlimited exploitation of nature by man disturbed the ecological balance between living and nonliving components of the biosphere. The adverse conditions created by man himself threatened the survival of not only man himself but also other living organisms.

Due to progress in industries, technology, chemicals, atomic energy etc., there are a number of industrial effluents and emissions of poisonous gases in the atmosphere and also added solid wastes which have lowered the quality of environment. The *pollution* is a necessary evil of all development. Due to lack of development of a culture of pollution control, there has resulted a heavy backlog of gaseous, liquid and solid pollution in our country. There is race in developed countries to exploit every

bit of natural resources to convert them into goods for their use and comfort and to export them to other needy countries. The industrialized countries dump lot of materials in their environment which becomes polluted. It would be no exaggeration to state that **Earth is a dying planet**. Unless concerted effort is made by all nations to protect the environment from the hazards that seems to increase with each passing day, serious disasters of a gigantic magnitude will overtake the human race.

In this lesson, environmental hazards and environmental pollution of different types have been discussed at length pointing also alongside the various measures to control them. Disaster management particularly with respect to earth quake, tsunami, landslides, flood, and cyclone have also been explained.

3:01 Environmental Degradation

3:01:1 Meaning of 'Environmental Degradation'

The term 'Environmental Degradation' is a wider and comprehensive concept, meaning the overall lowering of the environmental qualities because of adverse changes brought in by natural and human

activities in the basic structure of the components of the environment to such an extent that these adverse changes drastically affect all biological communities in general and human society in particular. Thus environmental degradation implies deterioration of the quality of the environment at the local, national and international levels.

3:01:2 Difference between Environmental 'Pollution' and 'Degradation'

The terms environmental **pollution** and environmental **degradation** are used interchangeably; but these are different concepts and have different meaning. Pollution is the cause for the degradation of environment. The *degradation* of environment is caused by pollution and *hazards / disasters*. The *hazards* or *disasters* are sudden by natural processes or by human activities which require the immediate relief and prediction can be made for the safeguard. The Tsunami that struck the Indian coasts on 26.12.2004 is a typical example of a disaster. These hazards cause sudden and severe pollution. Environmental pollution is taking place due to slow and gradual human activities, e.g., increase of human population, establishing factories and industries,

development of transportation facilities, etc. The pollution degrades the *quality of the environment* which can be protected by proper environmental management and assessment.

Environmental degradation leaves direct impact on the ecology and thus is caused ecological imbalance because of marked reduction in the ecosystem and ecological diversity. The ecological imbalance is the sign of environmental degradation; it is difficult to perceive environmental degradation directly at the initial stages. It can be easily observable in living organisms.

Thus *environmental degradation* is a wider concept than environmental pollution. It includes both natural processes and human activities while pollution is limited to lowering the quality of environment by human activities only.

3:01:3 Causes of Environmental Degradation

The following are the basic causes for the environmental degradation :

1. The development of modern technologies.
2. The increase in the human population.
3. High pressure on natural resources.

4. The high rate of exploitation of natural resources.
5. Growing industries and factories.
6. Increasing settlement of urbanization and
7. The development of economic functions of man.

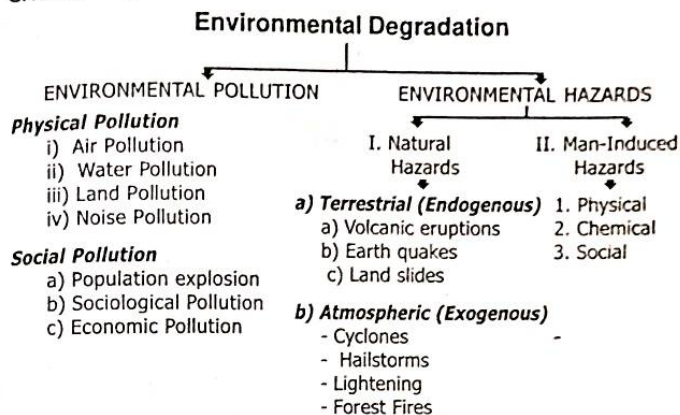
All these causes affecting the components of the environment are to be modified to such an extent that the adverse effects on the environment are reduced to minimum by 'Homeostatic Mechanism'. The physical and biological processes of the natural environmental system are self regulatory to compensate the negative feedback.

If the physical and biological processes cannot compensate and regulate the adverse effects on living organisms and man, it refers to '**Environmental Degradation**'. It cannot be set right by the self regulatory mechanism or homeostatic mechanism of the environment.

3:01:4 Types of Environmental Degradation

Environmental degradation is divided into two major categories - (i) Slow Degradation resulting from man-induced environmental pollution (ii) Extreme adverse changes in environmental quality brought

in immediately, by natural processes. The huge damage and loss to the quality of environment and living organisms are called 'Hazards' and 'Disasters'. Disasters include both man-induced hazards and natural hazards. Environmental degradation with its two major divisions are given in the following flow-chart.



3:02 Environmental Pollution

3:02:1 Meaning and Definition of Pollution

Pollution has been defined as the deliberate or accidental contamination of the environment with man's waste.

Pollution is matter in the wrong place.

Pollution is anything that is released into the environment which degrades it.

Pollution is presence of matter or energy in an unusual or unintended place.

Holister and Porteous in their Dictionary of the Environment give a very comprehensive definition as follows :

"A pollutant is a substance or effect which adversely alters the environment by changing the growth rate of species, interferes with the food chain, is toxic or interferes with health, comfort, amenities or property values of people."

Pollutants are the residues of things which we make use of and throw away. The rivers are polluted by wastes from factories, air by gases, thermal power plants etc. In fact, in countries where there have been the greatest technological and industrial advances, the worst pollution occurs.

3:02:2 Environmental Pollutants

The following are the main pollutants of environment in well developed and developing countries.

1. Deposited matter such as dust, smoke, tar, grit, soot etc.

2. Gases like Sulphur dioxide, Carbon dioxide, Carbon monoxide, Nitrogen Oxide, Hydrogen Sulphide, Ammonia, Fluorine, Chlorine, etc.
3. Chemical compounds such as aldehydes, arsenics, fluorides, detergents etc.
4. Metals like lead, iron, zinc, mercury etc.
5. Sewage
6. Radio-active substances
7. Agrochemicals like Fertilisers, Pesticides, Fungicides, Insecticides etc.
8. Photo Chemical Oxidants
9. Noise and heat

From the eco-system viewpoint, these pollutants can be classified into two basic types.

(i) Non-degradable pollutants

(ii) Biodegradable pollutants

3:02:2:01 Non-degradable Pollutants

There are materials and poisonous substances like aluminium cans, mercurial salts, long chain phenolic chemicals like plastics, polythene, PVC etc. and DDT that either do not degrade or degrade only very slowly in nature.

3:02:2:02 Biodegradable Pollutants

These are the domestic wastes like sewage, used paper cups etc. that can be rapidly decomposed under natural conditions. Problems arise with the biodegradable pollutants only when their input into the environment exceeds the decomposition or dispersal capacity i.e. when they accumulate.

3:03 Basis of Cost of Pollution

The following are the main bases for evaluating cost of pollution :

- i) Damage to crop production
- ii) Loss of resources by unnecessary wasteful exploitation
- iii) Medical care of health due to diseases
- iv) Soiling of buildings and textiles
- v) Corrosion of metals such as iron, steel, copper, brass, zinc, lead etc.
- vi) Pollution Control involving money, manpower etc. for dispersal of pollutants and for control devices developed.

3:04 Types of Environmental Pollution

Generally Pollution is classified either according

to the environment (air, water, land, light) in which it occurs or according to the pollutant (mercury, carbon dioxide, solid waste, noise, biocide, heat etc.) by which pollution is caused.

3:05 Land or Soil Pollution

Soil is very important environmental component for human, animals, and plants. It is the basic medium for food, vegetation and natural resources. It is also the base for the development of human culture and civilization. There are various aspects and components of soils - texture, structure, moisture, profiles and horizons.

3:05:1 Definition of Soil Pollution

The contamination of soil with excess of chemicals, fertilizers, insecticides, herbicides is known as soil pollution."

The decrease in the quality of soils either due to human activities or natural sources or by both is known as soil pollution or soil degradation. Soil pollution is caused due to soil erosion, decrease in plant nutrients, decrease in soil micro organisms, excess or deficit of moisture content, high fluctuation of temperature and lack of human content.

3:05:2 Sources of Soil Pollution

The major sources of soil pollution are :

- a) Deforestation
- b) Soil Erosion
- c) Soil Eruption
- d) Forest fires
- e) Excessive use of chemical fertilisers, pesticides, insecticides and herbicides
- f) Dumping of Industrial and Urban wastes
- g) Water-logging and seeping

3:05:3 Categorization of the Sources of Soil Pollution

The sources of soil pollution may be classified as under:

- i) **Physical sources:** e.g., soil erosion, volcanic eruption, forest fire, deforestation.
- ii) **Biological sources:** e.g., the micro-organisms, bacteria and protozoa.
- iii) **Air-born sources:** e.g., thermal power plants, radiations from industries and factories and nuclear radiations.

- iv) **Biocides sources:** e.g., chemical fertilizers and pesticides destroy micro-organisms, organic phosphate compounds.
- v) **Urban and industrial sources:** urban wastes such as paper, wood, plastic, polythene bags, packing materials, discharged sewage water etc. pollute the soil.

Among the industrial sources of soil pollution, the following are the important ones. (a) Paper Mills (b) Sugar Mills (c) Oil Refineries (d) Industries manufacturing Chemicals, Iron and Steel, Plastic Materials and Rubber Products, (e) Dyeing Units (f) Tanneries (g) Saw Mills and (h) Cotton Mills.

3:05:4 Adverse Effects of Soil Pollution

1. Soil pollutants directly affect the micro-organisms in the land.
2. Chemical fertilisers and pesticides, plastic materials and rubber destroy the organic phosphate compounds in the soil, due to which the soil loses its normal texture and capacity to retain water, resulting in the decreased quality and fertility of the soil.
3. Soil pollutants affect the ground water resources.

4. Soil pollutants lead to soil erosion which converts the fertile soil into waste land.
5. Soil pollution results in the decrease in agricultural production.
6. Crops growing in polluted soil become a cause for ill-health in human- beings.
7. Radiational wastes harm human and animal health.
8. Urban wastes are the eyesores, affecting health and hygiene.

3:05:5 Measures to Control Soil Pollution

It is very essential to control and prevent soil pollution because the existence of man, animals and vegetation depend upon the quality of soil. It is necessary to maintain and also increase the quality of soil. The essential commodities required for man, animals and plants come from soil. The following are the suggested measures to control soil pollution.

- i) The garbage and night soil can be converted into compost manures by cobalt and nickel techniques. Urban and industrial wastes could be safely disposed, not to reach the agricultural lands.

- ii) Avoiding as far as possible chemical fertilisers, pesticides and herbicides; resorting to organic farming using 'green manures' and 'bio-fertilisers'.
- iii) Restricting the use of D.D.T.
- iv) Industries emitting smoke and ashes are to be identified and monitored by pollution control agencies.
- v) Avoiding mono-culture and adopting crop rotation suitably.
- vi) Ensuring proper crop management and land use,
- vii) By encouraging recycling of wastes, solid wastes getting mixed with soil could be prevented to a large extent.
- viii) Reclamation of soil helps to reduce soil pollution.
- ix) Educating the farmers about the proper use of fertilisers and biocides.
- x) Developing awareness in the public about controlling soil pollution.

3:06 Water Pollution

Water is most important natural resource. It is vital for the maintenance of all forms of life and vegetations. We depend on water for irrigation,

industry, domestic needs, drinking purpose, for sanitation and disposal of waste. Our water bodies are ponds, lakes, sea, rivers, oceans which have become polluted due to industrial development and urbanization.

70% of the Earth is covered by water and that is why it is called the 'Blue Planet'. Only 2.5% of the world's water is fresh while 97.5% is found in our oceans. Only 0.3% of fresh water is available from rivers, lakes and reservoirs; 30% from ground water; the rest is stored in distant glaciers, icesheets and inaccessible areas.

In India, the available fresh water from both surface and ground water is 5 million -hectare metres and taking the average of 1500 cubic metres per year per person as the water need, the population of India will fall in the category of 'Water Stress'. At present ground water caters to about 5% of the irrigation requirements and 80% of domestic and sizeable portion of the industrial requirements. Day by day our water sources are getting increasingly polluted.

3:06:1 Meaning and Definition of Water Pollution

"The contamination of water with soluble sewage and industrial waste is called water pollution".

"Foreign materials either from natural and other sources contaminate water supplies and may be harmful to life, because of their toxicity, reduction of normal oxygen level of water, aesthetically unsuitable effects and spread of epidemics". - World Health Organisation (1966)

"Water pollution may be defined as alteration in physical, chemical and biological properties of water which may cause harmful effects on human and aquatic life".

Water pollution is defined as 'a natural or induced change in the quality of water which renders it unsuitable or dangerous as regards food, human and animal health, agriculture, fishing or leisure pursuits. - P Vivier (1958)

Water Pollution refers to :

- deterioration in physical, chemical and biological quality of water.
- causing the harmful effects on man, animals and vegetation and also quality of environment.
- deterioration in the quality of environment.

3:06:2 Sources of Water Pollution

There are two main sources of water pollutions as follows :

1) **Natural sources** : Such water pollutants include soil erosion, volcanic eruption, land slides, coastal and cliff erosion, floods, decomposition of plants and animals.

2) **Man-induced sources** : Of water pollutants include industrial development, urbanization, agricultural sources, cultural sources (*religious fairs and pilgrimage*). Kumbha fair at Allahabad is an example of cultural source of water pollution. Another example of cultural source of pollution is the immersion of Vinayaga Idols during Ganesh Chadurthi festival. The main sources of water pollution are:

- i) Sewage and other wastes in cities
- ii) Industrial effluents and waste products
- iii) Agricultural discharge (uses of chemical fertilizers and pesticides)
- iv) Thermal power plant and nuclear plant wastes
- v) Oil leakages and discharge

Oil leakages from ships or explosion of oil wells in the high sea cause spread of oil as a thick layer on the water surface which harms the aquatic plants and animals. Oil layers spread on the surface of sea water prevent sunlight to enter as well as air to dissolve in water, as a result of which sea based plants and animals die due to lack of oxygen.

3:06:3 Types of Water Pollution

Water pollution may be divided on the basis of storages of water such as :

- i) Surface water pollution
- ii) Lake water pollution
- iii) Ground water pollution
- iv) Sea water pollution and
- v) River water pollution

Water pollution is also classified on the basis of sources of water pollution e.g.,

- i) Sewage water pollution
- ii) Domestic waste pollution
- iii) Industrial waste water pollution, and
- iv) Solid waste water pollution

- v) Agricultural waste discharge (Chemical fertilizers, pesticides etc.)

Each type of water pollution affects the physical and biological components of *various aquatic systems* in different degrees and its ultimate effect on man remains quite drastic in medical, aesthetic and economical sense.

3:06:3:01 Sewage Water Pollution

The primary source of sewage pollution is the discharge of untreated sewage in water bodies, sometimes due to improper sewage - handling processes of municipal bodies. This is very common in major cities of the country.

Sewage is the waterborne waste derived from home (domestic waste) and animal or food processing plants. It includes human excreta, paper, cloth, soap, detergents, etc. These are a major proportion of the pollutants entering into our water bodies. There is uncontrolled dumping of wastes of rural areas, towns and cities into ponds, lakes, streams or rivers. Due to accumulation of sewage and other wastes in these bodies, they are not able to recycle them and their self-regulatory capability is lost. The decomposition of these wastes by aerobic microbes decreases due

to higher levels of pollution. The self-purifying ability of the water is lost and water becomes unfit for drinking and other domestic uses. Since decomposition of sewage and other waste is largely an aerobic process, accumulation of these in water increase its oxygen requirements.

Phosphates are the major ingredient of most detergents. They favour the luxuriant growth of algae which form water blooms. This extensive algal growth also consumes most of the available oxygen from water. This decrease in O_2 level becomes detrimental to growth of other organisms which produces a foul smell upon decay. Some decomposing plants are known to produce toxins such as *strychnine* which kills animals including cattle.

The following *methods* are to be used to check the water pollution through sewage waste :

a) The waste water must be treated before its discharge into lake or river. This would limit its nutrient input.

b) To stimulate bacterial multiplication in order to reduce the amount of nutrients solubilised in water. This would help disruption of algal food-web.

c) To check recycle of nutrients into the water through harvest and removal of algal blooms upon their death and decomposition.

d) To remove dissolved nutrients from water by physical or chemical methods. For instance, phosphorous can be removed by precipitation; nitrogen by biological nitrification and denitrification or by air stripping of NH_3 from an alkalised waste water, or by ion exchange, electrodialysis or reverse osmosis.

3:06:3:02 Water Pollution due to Industrial Effluents

A wide variety of both, *inorganic* and *organic pollutants* are present in effluents from breweries, tanneries, dyeing textiles, paper and pulp mills, steel industries, mining operations etc. The pollutants include oils, greases, plastics, plasticizers, metallic wastes, suspended solids, phenyls, toxins, acids, salts, dyes, cyanides, DDT, etc., many of which are not readily susceptible to degradation and thus cause serious pollution problems. Sulphuric acid waste from coal mines is a serious pollutant that increases the hardness of water, has disastrous effect on live organisms and corrodes concrete, etc., Nitrates,

Copper, Chromium, Cadmium, Mercury, Lead, etc., are the heavy metal effluents, discharged from industries.

3:06:3:03 Thermal Pollution of Water

The two chief pollutants are *heat* and *radioactive* substances. These are the wastes chiefly from power plants - *thermal* and *nuclear*, which use large quantities of water. Some other industries also give out waste water after use. Nuclear power plants are the source of radio nuclides.

The quantity of waste water is highest in the *thermal* power plants in the country. This waste water is returned after use at very high temperatures to the streams - a river or lake. This affects the aquatic life in these water bodies. This is also called *thermal pollution* since heat acts as a pollutant. Similarly, nuclear power plants besides causing fall out problems, also release waste heat. This also contributes to thermal pollution. Some plants and animals are killed outright by the very hot water. Though waste water from nuclear power plant is not very hot, but still has adverse effects on aquatic life. These are (i) early hatching of fish eggs, (ii) failure of trout eggs to hatch, (iii) failure of salmon to spawn,

(iv) increase in BOD, (v) change in diurnal and seasonal behaviour and metabolic responses of organisms, (vi) significant shift in algal forms and other organisms towards more heat-tolerant forms. This leads to decrease in species diversity (vii) affect changes in macrophytes and (viii) migration of some aquatic forms.

3:06:3:04 Water Pollution due to Agricultural Discharges

It is another kind of water pollution. These include chiefly the chemicals used as fertilisers and the pesticides (biocides) used in disease-control. Their discharges reach into the water bodies. As compared to developed nations, India has relatively a low use of these chemicals, thus discharges into water are still low. India uses about 16 kg/ha of fertilizers.

Modern agriculture rely heavily on a wide range of synthetic chemicals which include various types of fertilizers and biocides. They are dangerous, harmful and disturb the natural ecosystem. These chemicals affect the quality of the food stuffs. These chemicals make water unfit for drinking and also causes diseases.

Organisms which compete with man for their food and inflict heavy damage to the economy are called 'pests'. To eliminate the pests, chemical substances are used. Depending upon the organisms (pest) they kill, these chemical substances are called herbicides, insecticides or fungicides. Further, depending upon the nature of the chemicals contained in these biocides, they are also known as Arganochlorine, Argano Phosphorous, Carbonate and Nitro Phenol. Thus pesticides are the chemicals which find wide applications in modern agriculture, to control pests and save the crops. These biocides are harmful to man by entering into our food chain. Pesticides create various types of diseases related to kidney, blood, urine and brain tissues.

3:06:3:05 Other Sources of Water Pollution

- *Ground water pollution* by septic tanks, seepage pits etc.
- *Marine water pollution* - on their way rivers receive huge quantity of sewage, garbage, agricultural discharge, biocides and heavy metals.
- *Mercury pollution* - It is through industrial effluents. It is responsible for smallpox epidemic.

- *Lead pollution* - The main source of lead to water is the industrial effluents. It causes the damage to liver and kidney.
- *Fluoride pollution* - It is naturally present in water. The arid and semi-arid soils is due to fluoride contents. It causes the bone diseases.

3:06:4 Adverse Effects of Water Pollution

Water pollution affects in the following ways :

- i) Bacteria from sewage causes cholera, typhoid, hepatitis, dysentery etc. of water-borne diseases.
- ii) Nitrates contained in fertilisers, infiltrates in drinking water and inflict diseases like anemia.
- iii) Mercury from pesticides is toxic and causes foetal brain damage.
- iv) Chromium from industries is toxic and especially from tanneries is carcinogenic.
- v) Selenium from pesticides, industries causes cancer.
- vi) Photosynthesis get decreased in plants grown with polluted water.
- vii) Polluted water reduces significantly the life span of the crops.

3:06:5 Prevention and Control of Water Pollution

Control of water pollution requires several *remedial measures* and *Central Water Control Acts*. The following measures are suggested to control water pollution -

- 1) Maintaining stability of the eco-system
 - 2) Reutilization and recycling of waste
 - 3) Removal of pollutants and
 - 4) Management of water pollution.
- 1) *Maintaining Stability of the Ecosystem* : It is most scientific method to control and prevent water pollution. The basic principle is the reduction in waste input, thus water pollution is controlled at source. For example, by the following measures, contamination of water by the wastes could be controlled.
 - i) Domestic and industrial wastes should be permitted to be discharged to canals, ponds, rivers or the sea only after duly treating them to safe limit.
 - ii) Water pollution could be controlled to a large extent, if appropriate legal measures are strictly enforced.

Several methods of biological as well as physical can be used to restore species diversity and to maintain ecological balance in the water bodies.

- 2) *Reutilization and Recycling of Waste* : The various types of wastes include industrial effluents and thermal pollutants which may be reutilized. The urban waste is reused to generate cheaper fuel gas and electricity. Efforts are being made in the development of suitable technology for waste water so that it can be utilized in agriculture and prepare some useful products.
- 3) *Removal of Water Pollutants* : There are various types of pollutants present in water bodies which can be removed by suitable methods, e.g., absorption, electrodialysis, ion exchange, reverse osmosis, etc. This process is used for purifying water from sewage.

The *solar power* can also be used for cleaning up polluted water cheaply. Such type of experiments are being conducted in our country and also in Western countries.

3:07 Air Pollution

3:07:1 Meaning and Definition of Air Pollution

Due to some natural processes like volcanic

eruptions, deflation of sand and dust etc. or human activities when the amount of solid waste or concentration of gases other than oxygen (O_2) increases in the air, which normally has constant percentages of different gases in it, the air is said to be polluted. The contamination of air with smoke, dust, and harmful gases is called air pollution.

The atmosphere is a gaseous envelope which surrounds the earth from all sides and the air is a composition of several gas mainly nitrogen (78.4%), Oxygen (20%), Argon and other gases (1.5%) and Carbon dioxide (.0314%). Air is very important for all types of life in the biosphere. Human life is not possible without air because man can survive for a few weeks without food, and for a few days without water but cannot live even for a few minutes without air.

3:07:2 Sources of Air Pollution

There are two major sources of air pollution as follows :

- (1) *Natural Sources* : Such as volcanic eruptions, deflation of sand and dust, forest wild fires, natural vegetation, etc.,

- (2) *Man-made sources* : Or human activities. Such as industries, factories, urban centers, aircraft, nuclear experiments, automobiles, agriculture, power plants etc.

From the different sources of air pollution, a variety of pollutants are released into atmosphere. The principal air pollutants emitted from these different sources are as follows :

1. *Carbon compounds* : These are mainly carbon dioxide and carbon monoxide, the former released by complete combustion of fossil fuels and the latter by automobile exhausts.
2. *Sulphur compounds* : These include Sulphur dioxide (SO_2), Hydrogen Sulphide (H_2S) and Sulphuric Acid mostly released by fossil fuel (coal, etc.) based power generating plants (thermal plants) and industrial units like refineries.
3. *Nitrogen Oxides* : These include chiefly nitrogen monoxide, nitrogen oxide and nitric acid mostly released by automobiles, power plants and industries.
4. *Fluorocarbons* : These chemicals coming from industries, insecticides spray, etc. affect the ozone layer and result in the process of global warming.

5. *Hydrocarbons* : These are chiefly benzene, benzpyrene, etc., which are mostly discharged by automobiles and industries.
6. *Metals* : These include chiefly lead, nickel, arsenic, beryllium, tin, vanadium, titanium, cadmium, etc., present in air as solid particles or liquid droplets or gases. They are produced mostly by metallurgical processes, automobiles, sea spray etc.

3:07:3 Types of Air Pollution

Air pollution is classified on two bases as follows :

1. On the basis of *types of pollutants*, and
 2. On the basis of *types of sources of pollutants*.
1. On the basis of types of pollutants air pollution can be further divided into two categories –
(a) *Gaseous* pollution and (b) *Particulate* air pollution.
 2. Air Pollution, on the basis of sources of pollution is sub- divided into six categories :
(a) *Automobiles* pollution (b) *Industrial* pollution.
(c) *Thermal* pollution (d) *Urban* pollution
(e) *Rural* pollution and (f) *Nuclear* pollution.

Generally these approaches are used to describe the air pollution. There is overlapping among the various categories.

3:07:3:01 Automobile Air Pollution

During this period of technological development, there is very rapid increase of cars, trucks, buses and two wheelers. It has raised the traffic density in our towns and cities. The vehicles create tonnes of gaseous pollutants into the air daily. The brief idea of gaseous pollutants has been given in the following paragraph.

The toxic vehicular exhausts are a source of considerable air pollution, next only to thermal power plants. The ever increasing vehicular traffic density posed continued threat to the ambient air quality. There are over 1400 million cars, trucks and buses all over the world. In India there are over 300 million vehicles, of which more than 65% are two-wheelers operating on petrol. In all the major cities of the country about 800 to 1000 tonnes of pollutants are being emitted into the air daily, of which 50% come from automobile exhausts. In the major metropolitan cities, vehicular exhaust accounts for 70% of all Carbon Monoxide (CO), 50% of all hydrocarbons,

30-40% of all oxides and 30% all Suspended Particulate Matter (SPM). It is estimated that a car (without cleaning device), on burning 1000 litres of fuel emits 350 kgs. CO, 0.6kg of SO₂, 0.1kg of lead and 1.5 kg of SPM.

3:07:3:02 Industrial Air Pollution

India is the agricultural country but there is rapid development in the agricultural technology and engineering. The economic development is based on both agricultural and industrial development of nation. There is also rapid industrial development in our country. There are industrial towns and complexes in our big cities.

There are a number of industries which are source of air pollution. Petroleum refineries are the major source of gaseous pollutants. The chief gases are SO₂ and NO₂. *Mathura-based petroleum refinery is posing threat to Taj Mahal in Agra and other monuments at Fatehpur Sikri complex.* Cement factories emit plenty of dust, which is a potential health hazard. Stone crushers and hot mix plants also create a menace. The SPM levels in such areas of stone crushing are more than five times the industrial safety limits. There are many food and fertilizer industries which emit gaseous pollutants.

Stubble burning by the farmers also pollutes the air and bring down the quality of the air to a greater extent. There are also chemical manufacturing industries which emit acid vapours in air. During last 10 years the number of industrial units in Delhi, increased from about 20,000 to 55,000. Nearly 40,000 of these are located in predominantly residential areas. The worst industrial air pollution recorded so far in India is the **Bhopal Gas Tragedy** in which on December 3, 1984 about 40,000 Kg of deadly methyl isocyanide gas from the Union Carbide Plant leaked into air, killing more than 2000 people and causing serious and permanent injuries to thousands of people. The adverse effects of this tragedy still persists.

3:07:3:03 Thermal Air Pollution

There are a number of thermal power stations and super thermal power stations in our country. The National Thermal Power Corporation (NTPC) has set up several coal-powered power stations to augment the energy generation. These are located in the States of Madhya Pradesh, Uttar Pradesh, Odisha, Chattisgarh, Andhra Pradesh and West Bengal. The coal consumption of thermal plants is several million tonnes. The main pollutants are fly ash, sulphur oxide and other gases and hydrocarbons.

3:07:3:04 Nuclear Air Pollution

The importance of the nuclear power industry makes it essential in any discussion of air pollution that we devote some attention to nuclear pollution. The four areas of concern in any nuclear power plant are radiation releases from the plants themselves and the nuclear fuel reprocessing plants, the possibility of accident, the disposal of radioactive wastes and the radiation problem associated with the mining of the nuclear fuel.

One of the classical cases of nuclear accidents is the **Chernobyl Atomic Power Plant accident** in Russia on April 26, 1986. The explosion that took place in this nuclear plant resulted in innumerable deaths, besides incapacitating thousands of people. The radiation effects of the accident were not confined to Russia alone. It is reported to have affected the neighbouring countries too. About a week after the accident, nuclear clouds could be seen in Eastern Europe and other places. Many people died of nausea and cerebral haemorrhagia. Such is the devastating effect of nuclear pollution. Another major nuclear accident was Fukushima nuclear disaster in Japan on 11th March 2011.

3:07:4 Adverse Effects of Air Pollution

Air pollution causes several kinds of damages to man, animals, plants and also on weather, climate and atmospheric processes. Some significant effects have been enumerated here in the following paragraphs.

- 1) Air pollution has very serious types of pathological effect on man. It causes several types of diseases to man.
- 2) It causes obstruction to vision of pilots and has caused air craft accidents.
- 3) Air pollution has damaged both agricultural crops and natural plants and vegetation.
- 4) There has been injury to agricultural livestocks, particularly from air borne fluoride and arsenic pollution.
- 5) Air pollution also takes its toll of buildings and other man made objects such as famous buildings, monuments and art treasures.
- 6) Air pollution also affects weather and climate on regional, continental or global basis. Acid rain and ozone depletion caused by air pollution cause severe damages to man and other living organisms.

- 7) The increased amount of carbon dioxide in atmosphere, causes increase in the temperature of earth. It is known by the phrase '**Green house effect**'. It also tends to prevent the long wave radiations from earth. Thus it has atmospheric effect. The intensity of *radiation* will cause skin cancer.
- 8) The jet engines of supersonic aircraft flying at high attitude release nitrogen oxide which destroy ozone molecules. It has also atmospheric adverse effects.

3:07:5 Control and Prevention of Air Pollution

The several types of air pollutions can be controlled by employing modern technology and scientific researches; but the costs to be borne by the public in the form of higher prices for manufactured goods, higher taxes, reduction of profit margins in industry and more restriction on human activities. It involves the benefits to improve the quality of the environment and human health, agriculture and vegetations. It will also improve the ecological balance and stability of ecosystem.

The main *sources* of air pollution are -
1) Automobiles, 2) Industrial development, 3) Power plants - coal based and thermal power plants. Thus

steps are to be taken to control air pollution at source and after the release of pollutants.

1) *Automobiles Air Pollution* : The following are some of the measures to be followed for controlling air pollution by automobiles.

- a) To check pollutants emission from vehicular exhaust.
- b) To control evaporation from fuel tank and carburettor.
- c) To use the filters, and
- d) To control through law –Motor Vehicles Act 1988 and other Amendments thereon.

2) *Industrial Air Pollution* : Air pollution by industry and power plants waste can be checked by devising measures for removal of the particulate matter and gaseous pollutants from the wastes. The three types of equipments or devices used are -

- a) Cyclone collectors
- b) Electrostatic precipitators and
- c) Control through law – Acts of industry

3) *Gaseous Pollutants* : The following four methods can be used to control air pollution from gaseous pollutants :

- a) *Wet systems* : The alkali liquid reacts with sulphur dioxide to produce a precipitate.
- b) *Dry Systems* : Water in contact with sulphur dioxide produces sulphuric acid.
- c) *Wet dry systems* : This method is very effective in dry cleaning plants, printing shops, paint factories, food processing plants, etc.
- d) *Control through Law* : Acts for Industries are to be enforced strictly to prevent gaseous pollution.

3:07:6 Role of Education in Controlling Air Pollution

Education is a powerful instrument to control social problems. Air pollution is a serious social problem of these days. Everyone is concerned and affected by the problem. The air pollution and noise pollution are interrelated; thus both are included under Air and Noise Pollution Prevention Act. The legal steps have already been taken by state and central governments, to prevent and control the air pollutions which have been discussed in earlier paragraphs.

Education can also play a significant role in the prevention and controlling of air pollution. It is very essential to provide the awareness and develop consciousness among students about air pollution and its resources and ways. Formal, non-formal and adult education programmes should provide awareness and develop feelings and attitudes or consciousness about the problem which has the serious adverse effects on human health. Charts and mottos are to be prepared on air pollutions — resources and remedial measures. Now a days media and newspapers are doing the same, e.g., 'Save the Oil', 'protect the air'. The students should also be given this type of knowledge. The oil resources are limited and decreasing rapidly, thus the day is not far away that automobiles and aeroplanes will not be able to make use of oil resources.

3:08 Marine Pollution

Marine pollution is defined as the discharge of the wastes and sewage into the sea resulting in harm to living organisms, hazards to human health, hindrance to fishery and impairment of quality of sea water for use. Marine pollution is associated with the change in physical, chemical and biological properties of the sea water.

It is wrong to consider that discharging of wastes into the sea is harmless. Marine organisms get seriously affected by the pollution of sea water. Toxic chemicals in fish enter food chain and affect human health. Anything that affects the quality of sea water is considered as a marine pollutant (e.g. solid wastes, sewage water, industrial discharges spread of oil on the sea water from leaking containers in the ships etc.)

3:08:1 Sources of Marine Pollution

1. The commonest pollutant of the sea water is raw or treated sewage. It is interesting to note that in reasonable quantities sewage can be harmless or beneficial. But, over burdening with nutrients give rise to eutrophication.
2. Synthetic detergents which have widely been used now as a replacement of soap is the second most predominant pollutant of sea water. Sodium tripolyphosphate contained in detergent soap / powder makes water alkaline.
3. When sewage discharged in sea, contains chemicals such as fertilisers, pesticides, heavy metals like nickel and lead or D.D.T. (use of which is banned in developed countries) serious sea pollution occurs.

4. Ships carrying crude oils when collide with other ships or rocks sink in the sea, causing oil spread over a vast surface area on the sea, which pollutes the sea water, as a result of which marine plants and animals perish due to lack of oxygen and sunlight.

We can control oil leaks in sea but have no control over the destruction of marine life due to oil leak.

3:08:2 Control of Marine Pollution

To control the marine pollution the following measures could be undertaken :

- i) Exhaust the oil from the sinking oil carriers (Ships) and help the oilslicks to reach the seashore by the action of tides.
- ii) Burn the thick layers of oilslicks.
- iii) Sinking oilships, could be made to immerse deep into the sea.
- iv) Emulsification of oilslicks to make them harmless.
- v) Preventing the discharge of sewage into the sea, at the point of generation itself.
- vi) Implementing ruthlessly the provisions of law pertaining to marine pollution.

3:08:3 Adverse Effects of Marine Pollution on Eco-System

The following are the adverse effects of marine pollution on eco-system.

1. Coral reefs, the natural habitats of marine life gets affected by excessive discharge of sewage, oilslicks, hot water and metal particles and as a result of which marine species do not breed and get multiplied.
2. The solid and liquid wastes as well as oilslicks affect the plants and fishes in the sea.
3. Solid wastes discharged into the sea, block the penetration of sunlight into the sea water, and cripple photosynthesis.
4. Heavy metals and radioactive substances through 'biomagnification' pollute sea food. Health of those who take these sea food gets affected.
5. As oilslicks greases the bodies of birds and mammals, their movements are significantly curtailed as a result of which they become sick and die.

3:09 Noise Pollution

3:09:1 Meaning and Definition of 'Noise Pollution'

Noise, defined as the unwanted high intensity sound, is an important type of urban pollution that is capable of not only causing annoyance and hearing loss but also in some cases adverse physiological and psychological effects.

3:09:2 Sources of Noise Pollution

Sources of noise could be broadly classified as

- a) Natural Sources
- b) Man-related Sources
 - a) *Natural Sources* : These includes thunder bolts, noisy heavy downpour, whistling storms, icy rainfall etc.
 - b) *Man made Sources* : These include industrial noise, transport noise, neighbourhood noise and noise from construction work. Noise Pollution is included in air pollution.

We shall discuss briefly about each of these categories.

1. Transport Noise

Main menace of noise, comes from transport. It mainly includes the following :

- (i) Engine noise (ii) Noise from exhaust pipe
- (iii) Noise due to banging of car doors (iv) Noise due to sudden brake (v) Noise due to horn.

Noise due to road traffic could be controlled by traffic diversions, implementing the schemes like one way traffic in narrow roads, developing 'by-pass' roads and 'ring roads', separate lane for cars and buses, putting speed breakers etc. Rail traffic noise could be controlled by allowing passengers alone on the rail platforms and locating the rail stations on the outskirts of the towns and cities where the density of population is low.

2. Aircraft Noise

Jet aircrafts generate more noise than other types of aircrafts. Aircraft noise could be controlled by fixing noise suppressors; but this makes the air craft heavier and thereby increasing the consumption of fuel. New take-off strategies are to be tried soon, to reduce noise pollution.

3. Noise from Construction Work

Noise arises due to handling of tools, equipments and techniques used during the construction of houses and apartments, dams, industrial plants and sheds. Modern techniques and tools are to be employed to reduce noise due to construction work.

4. Industrial Noise

Frictions, vibrations, air ejections, etc. that take place during the operation of machineries in industries produce noise. This could be considerably reduced by the application of lubricants, arranging separate path ways for incoming and outgoing air in machines, adopting better design for engine and other parts of machinery etc.

3:09:3 Noise Limits

Our country has the distinction of having three of the noisiest cities in the world - Mumbai, Kolkatta and Delhi. In Mumbai on non-festive days the noise level is 70 dB, more than the WHO prescribed standard of 55 dB, during the day and 45 dB, at night.

On festive days it increases to 90 dB. Many people work and live in noisy conditions in our cities. The following table gives maximum noise levels at some busy places.

Noise Levels at Different Locations & Situations

Situation	dB
Road traffic near residential areas	70
Noise at building sites	70
Factory work for an 8-hour day	90
Indiscriminate use of loudspeakers	100
Bursting of crackers	128
Eardrum rupture	180
Near airports	78-116
Near railway stations	74-98
Near bus terminus	68-91
Train noise - inside II class compartments with window shutters open	86
Fast trains while crossing platforms	90

3:09:4 Effects of Noise Pollution

Medical authorities feel that constant occupational exposure to noise levels of 90 decibels in the range of human hearing are dangerous. Short exposures to high intensity noise produce temporary

hearing loss; longer exposures will lead to permanent hearing loss. Noise also produces annoyance in human beings and can affect proper rest and sleep. Noise can affect verbal communication and concentration to work. Though noise upto 60 decibels does not interfere with speech, higher levels of noise force people to raise their voices. While noise levels of 85 decibels make understanding barely possible with loud shouts, noise levels beyond 90dB make understanding virtually impossible. Noise also interferes with tasks requiring a high degree of concentration, such as office work and learning in schools.

Noise also affects the physiological function of the human body. One important effect is constriction of the small blood vessels, resulting in the reduction of blood-flow. Other physiological effects include paling of the skin, dilation of the pupils of the eyes, decrease of gastric secretion, tensing of voluntary and involuntary muscles, increase in diastolic blood pressure, nervousness, irritability and anxiety.

As noise reduces the depth and quality of sleep, it affects one's health. Though noise does not make a person awake from sleep, it does alter his state of sleep. Without a sound sleep, one may feel mental

fatigue and physically tired, which affects the efficiency of performance.

3:09:5 Control of Noise Pollution

- 1) **Source Control** : Gadgets (Silencer and devices) are to be developed to control or reduce noise at source. Redesigning the noisy machines and manufacturing process could be undertaken.
- 2) **Transmission Control** : The room walls can be covered with sound absorbers, providing shields and barriers.
- 3) **To protect exposed persons or workers** : Devices as ear plugs and ear muffs can be provided to workers of industries.
- 4) **Plantation or vegetation** : Trees should be planted along high ways, streets and other noisy places like schools, bus terminus, hospitals and factories; big trees such as neem, coconut etc. are good for this purpose.
- 5) **Through Acts or Law** : The noise pollution has now been included under Air Pollution Control Act. Measures like reducing the sound produced by the automobile horns, forbidding the use of horns in certain places like hospitals, schools etc. and regulating the use of loud

speakers in religious, domestic and political functions should be strictly implemented.

- 6) A rational town planning can help in reducing out door noise pollution.
- 7) **Education** : By educating students and the public about noise nuisance through news media.

3:10 Thermal Pollution

Thermal pollution refers to the detrimental effects of hot effluents discharged by various power plants and industries. It denotes the impairment of quality and deterioration of aquatic and terrestrial environment.

Industrial plants like atomic and coal fired power plants, oil field generators, factories and mills utilise water for cooling purpose. The heated effluents (waste water coming out) are discharged at a temperature 5 to 10°C higher than the temperature of the intake waters. Hot effluents discharged in rivers, lakes, streams, ponds etc. reduce the concentration of dissolved oxygen which produces deleterious effects on the aquatic biots and ecosystem.

The accelerated pace of development, rapid industrialisation and urbanisation have increased the demand for power as well as industries to produce goods in a massive scale. Human activities, today are constantly adding pollutants to air and water at an alarming rate. Following are the important sources of thermal pollution - (i) Atomic power plants (ii) Coal-fired power plants (iii) Hydro - electric power plants (iv) Industrial effluents and (v) Domestic sewage.

3:10:1 Adverse Effects of Thermal Pollution

1. Reduction in Dissolved Oxygen

Concentration of dissolved oxygen decreases with increase in temperature and as such the hot industrial effluents discharged in streams and rivers destroy the aquatic organisms due to non-availability of adequate oxygen.

2. Change in Water Properties

A rise in temperature changes the physical and chemical properties. The decrease in solubility of gases, viscosity and density increases the settling speed of suspended particles which seriously affect the food supplies of aquatic organisms.

3. Interference with Biological Activities and Reproduction

Rise in temperature increases the toxicity of poisonous substances present in water which makes fishes to perish. Changes in temperature upsets the biological activities of aquatic organisms such as metabolism, respiration, excretion, reproduction, etc. Warm water disturbs spawning and also destroys the laid eggs of fish.

4. Invasion of Destructive Organisms

Thermal pollutants may permit the invasion of organisms that are tolerant to warm water and highly destructive. Hot water causes bacterial diseases in fishes.

5. Change in Algae

Excessive nutrients from the washout waters from lands, combined with thermal pollution cause an excessive growth of algae like diatoms, green and blue-green algae. This affects the aquatic food chain and thereby increase in toxicity to fishes and cut short their longevity.

3:10:2 Measures to Reduce Thermal Pollution

The following are some of the important measures to control thermal pollution of industries.

- i) Letting out the hot industrial effluents into artificial ponds and allow them to cool before discharging into rivers, streams and lakes.
- ii) Arranging for cooling towers in large industries.
- iii) Channeling of thermal effluents to power generating plants.

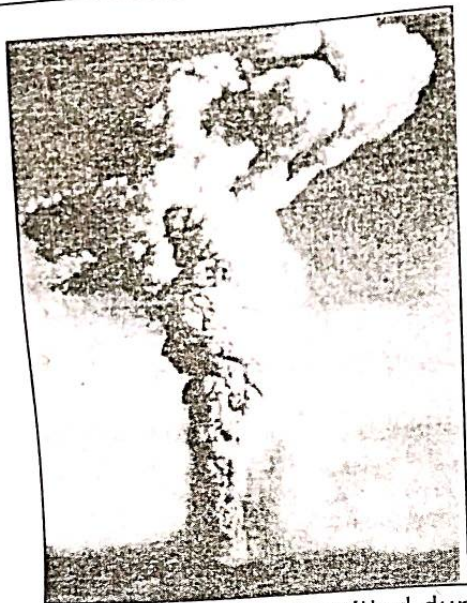
3:11 Radiation Pollution

Today, with the existence of nuclear power plants, the presence of radioactive fallout in the atmosphere has increased. The use of radiation and nuclear devices for constructive purposes have also been responsible for rapidly increasing the radiation pollution. A few of these radio active pollutants are Argon, Cobalt, Krypton, Cerium, Strontium, Uranium and Plutonium.

3:11:1 Radiation Sources

The radiation sources are either natural or man-made.

Natural radiation is produced by naturally occurring radioactive elements. The amount of whole



Enormous Radiational dust emitted during the Hiroshima bombing

body radiation produced by these naturally occurring radioactive elements varies from place to place depending upon the nature of the soil and the distance to mineral deposits. Cosmic radiation is another important source of natural radiation.

Among the man-made sources of radiation are the medical procedures. An average chest X-ray gives 200-mrem (a radiation unit) exposure to the part of the body that is exposed to radiation. The average gastrointestinal tract examination gives 22 rem (a radiation unit). The contribution of nuclear power

plants to radiation pollution is, however, quite small because of the safety standards set by the authorities. Radioactive fallout of nuclear weapon tests is another major source of radiation pollution.

3:11:2 Effects of Radiation Pollution

The effects of radiation can be categorized as

- ▲ Direct Effects i.e. fragmentation of biological molecules like DNA in the cell nucleus.
- ▲ Indirect Effects i.e. fragmentation of biologically less vital molecules like water with the formation of free radicals that can affect more important molecules and impair their effectiveness.

From the biological point of view the effects can also be classified as

- ▲ Somatic Effects i.e. the effects on the body itself being of direct concern to the person exposed to the radiation.
- ▲ Genetic Effects i.e. the effects that involve mutations of the chromosomes or genes in sex cells, posing a potential hazard to the descendants of the person exposed to the radiation. This is of concern to the whole society.

The estimated short-term effects of single dose, whole body radiation exposures in man are as follows:

Rad is one of the units used to measure radiation.

Below 25 rads	No effect that is observable
25-50 rads	Effect is detectable from this limit
50-100 rads	Slight temporary blood changes
100-250 rads	Nausea, vomiting and fatigue
250-500 rads	May prove fatal
500-1000 rads	50 per cent of the victims may die
Above 1000 rads	No victim would survive

Exposure to a few hundred rads often leads to acute radiation illness, with symptoms like nausea, fatigue and vomiting, followed by decrease in erythrocytes (Red Blood Cells) and leucocytes (White Blood Cells). Subsequently the victim suffers from anemia and becomes susceptible to bacterial infection and hemorrhaging for a brief period. This condition often ends in death.

3:11:3 Reduction of Nuclear Pollution

- i) Reducing the number of nuclear power plants in a phased manner. (It is worth recalling here that Japan leads the way by shutting down its nuclear power plants one after the other).
- ii) Removing and storing of radio active wastes with high safety and security must be ensured.

- iii) Nuclear pollution could be controlled by implementing multi level safeguarding systems employing advanced techniques and devices.
- iv) Safety measures are to be implemented with determination allowing no excuse or compromise; efforts are to be taken to reduce considerably the radiational emissions from the plant.

3:12 Light Pollution

3:12:1 Concept of Light Pollution

Light pollution, also known as **Photo-pollution** or **luminous pollution**, is excessive, misdirected or obtrusive artificial light. Pollution is the 'adding of' or 'added light itself', in analogy to added sound, carbondioxide etc. Adverse consequences are multiple; some of them may not be known yet.

Many definitions are given for 'light pollution', of them following four are widely accepted. Light pollution is :

- i) degradation of photic habitat by artificial light
- ii) alteration of natural light levels in the outdoor environment owing to artificial light sources.
- iii) alteration of light levels in the outdoor environment (from those present naturally) due

to man-made sources of light. Indoor light pollution is such alteration of light levels in the indoor environment due to sources of light, which compromises human health.

- iv) introduction by humans, directly or indirectly, of artificial light into the environment.

The first three of the above four definitions, describe the state of the environment. The fourth (and the newest) one describes the process of pollution by light. Thus we can say **all intruding light which affects the natural light level of a place are considered as a photo-pollute.**

3:12:2 Sources of Light Pollution

Light pollution is a side effect of industrial civilisation. Its sources include :

- i) lighting arrangements done in building exterior and interior.
- ii) illuminating advertisement boards
- iii) using glow lights in commercial complexes
- iv) powerful lights used in offices and factories
- v) powerful street lights
- vi) illuminated sporting venues and cinema shooting spots / studios.

Generally light pollution is highly prevalent in industrially developed countries as well as those with high levels of population.

3:12:3 Adverse Effects of Light Pollution

- i) Light pollution competes with star light in the night sky and interferes with astronomical observation.
- ii) Light pollution disrupts eco-systems.
- iii) It affects the sleep of people, makes them irritated and feel mental and physical fatigue.
- iv) Electricity is needlessly wasted.

3:12:4 Types of Light Pollution

Light Pollution is divided into two main types :

- i) Unpleasant light that intrudes on an otherwise natural or low-light setting.
- ii) Excessive light (generally indoors) that leads to discomfort and adverse health effects.

3:12:4:01 Light Trespass

Light trespass occurs when unwanted light enters one's property, for instance, by shining over a neighbour's fence. A common light trespass problem occurs when a strong light enters the window of one's home from the outside, causing problems such as

sleep deprivation or the blocking of an evening view.

3:12:4:02 Over - Illumination

Over-illumination is excessive use of light. [(e.g.) illuminated advertisement boards, illuminated sporting venues, cinema studios, factories, outdoor decorations etc.] This results in the enormous wasting of valuable electric energy. An estimate reveals that out of total consumption of energy in the U.S. equivalent 18.8 million barrels of petroleum per day, approximately two million barrels of oil is wasted in over-illumination.

Reasons for Over-illumination

Over-illumination stems from several factors :

- ▶ Not using timers, occupancy sensors or other controls to extinguish lighting when not needed;
- ▶ Improper design, especially of workplace spaces, by specifying higher levels of light than needed for a given task;
- ▶ Incorrect choice of fixtures of light bulbs, which do not direct light into areas as needed;
- ▶ Improper selection of hardware to utilize more energy than needed to accomplish the lighting task;

- ▶ Incomplete training of building managers and occupants to use lighting systems efficiently;
- ▶ Inadequate lighting maintenance resulting in increased stray light and energy costs;
- ▶ "Daylight lighting" demanded by citizens to reduce crime or by shop owners to attract customers;
- ▶ Substitution of old mercury lamps with more efficient sodium or metal halide lamps using the same electrical power; and
- ▶ Indirect lighting techniques, such as lighting a vertical wall to bounce photons on the ground.

Most of these issues can be readily corrected with available, inexpensive technology, and with resolution of landlord / tenant practices that create barriers to rapid correction of these matters. Most importantly public awareness would need to improve for industrialized countries to realize the large payoff for over-illumination.

3:12:4:03 Glare

Glare can be categorized into different types. One such classification is described in a book by **Bob Mizon**, coordinator for the British Astronomical Association's Campaign for Dark Skies. According to

this classification:

- i) Binding glare describes effects such as that caused by staring into the Sun. It is completely blinding and leaves temporary or permanent vision deficiencies.
- ii) Disability glare describes effects such as being blinded by oncoming car lights, or light scattering in fog or in the eye, reducing contrast, as well as reflections from print and other dark areas that render them bright, with significant reduction in sight capabilities.
- iii) Discomfort glare does not typically cause a dangerous situation in itself, though it is annoying and irritating at best. It can potentially cause fatigue if experienced over extended periods.

Glare from bad lighting is a public-health hazard, especially for the old people. Glare light scattering in the eye causes loss of contrast and leads to unsafe driving conditions, much like the glare on a dirty windshield from low-angles sunlight or the high beams from an oncoming car; in essence bright and / or badly shielded lights around roads can partially blind drivers or pedestrians and contribute to accidents.

3:12:4:04 Light Clutter

Light clutter refers to excessive groupings of lights. Groupings of lights may generate confusion, distract from obstacles (including those that they may be intended to illuminate), and potentially cause accidents. Clutter is particularly noticeable on roads where the street lights are badly designed, or where brightly lit advertising surrounds the roadways. Depending on the motives of the person or organization that installed the lights, their placement and design can even be intended to distract drivers, and can contribute to accidents.

Clutter may also present a hazard in the aviation environment.

3:12:4:05 Skyglow

Skyglow refers to the glow effect that can be seen over populated areas. It is the combination of all light reflected from what it has illuminated escaping up into the sky and from all of the badly directed light in that area that also escapes into the sky, being scattered (redirected) by the atmosphere back toward the ground. This scattering is very strongly related to the wavelength of the light when the air is very clear. **Rayleigh Scattering** dominates in such

clear air, making the sky appear blue in the daytime. Skyglow is of a particular irritation to astronomers because it reduces contrast in the night sky to the extent where it may even become impossible to see anything but the bright stars.

3:12:5 Control of Light Pollution

Disputes are common when deciding appropriate action, and differences in opinion over what light is considered reasonable, and who should be responsible. Authorities have also taken a variety of measures for dealing with light pollution, depending on the interests, beliefs and understandings of the society involved. Measures range from doing nothing at all, to implementing strict laws and regulations about how lights may be installed and used.

3:13 Solid Waste Pollution and Management

Solid waste substance are those materials which become useless and hence waste after short period of their use, such as news papers, bottles, plastic bags, polythene, packing materials and garbages. The solid wastes are those that are discarded after their uses. The *environmental pollution* is caused by these solid wastes which are increasing with rapid rate of urbanization and industrial development. The main problem is the disposal of solid wastes.

3:13:1 Sources of Solid Wastes

The solid waste materials may be classified into two categories. Production of solid waste materials

- 1) at production centres.
- 2) at consumption centres after the use.

The production centres, both small and large scale industries, produce the solid waste materials.

3:13:2 Categories of Solid Wastes

The consumption centres use and throw the leftouts as waste materials. Types of Solid Waste Substances : These types of pollutants are divided into several types based on their sources.

- 1) Mining wastes 2) Agricultural wastes
- 3) Industrial wastes 4) Municipal wastes 5) Packing wastes 6) Human wastes 7) Animal wastes and
- 8) Radioactive waste.

1. Domestic Wastes

Most of the households are swept clean daily and the solid wastes like left out food, bottles, papers, plastic bags, containers, polythene bags, worn out clothes, cadrboard boxes, packing materials such as thermocoal boards, leaf plates, wrappers, fruit peels, non-biodegradable cups etc. are thrown over the

compound wall onto the backlane or on the street. There, they pile up, rot, stink and breed flies, mosquitoes and bacteria, causing pollution and diseases.

2. Municipal Wastes

This includes discarded materials from houses, shops, commercial establishments, business complexes, eateries, roadside stalls, etc. collected daily by the municipal sanitary staff as well as garbages, leaves and plants thrown out as garden wastes etc. that are removed once in a week by municipal authorities.

3. Construction Wastes

Builders, contractors and house owners deposit the debris and other construction wastes just outside the houses and buildings on the roads, which narrows down the carriage way for the vehicles causing obstruction to traffic.

4. Industrial Wastes

Used up packing materials from the business houses and industries, wastes emerging out from textile, paint and chemical industries, steel plant wastes, (slags) plastic and rubber wastes, wastes emerging out from food preservation industries, metal

scraps etc. are industrial wastes which produce severe pollution. Some of the industrial wastes are explosive and inflammable (those from match stick and fire cracker manufacturing factories); some are toxic (waste from petro-chemical industries, drug and fertilizer manufacturing units, sugarmills etc.); and some others are radiating (wastes from atomic power plants, hospital wastes etc.) It is reported that every year more than 5 lakh tonnes of industrial wastes are generated in India. Most industrial solid wastes are disposed of in an unscientific manner surreptitiously on open plots or in water bodies in urban areas. Absence of adequate disposal facilities for the industries is a compelling factor for indiscriminate disposal of industrial solid wastes.

5. Agricultural Wastes

Cattleshed wastes and crop residues are the main agricultural wastes. They include rice bran, rice husk and rice-straw, left outs from crushed sugar canes, tea and coffee waste, tobacco waste, coconut waste, rubber waste, jute waste etc. Human excretions also are included in this category.

6. Mining Wastes

"Tailings", slag heaps, culm piles at coal mines

etc. are the mining wastes which cause high pollution and health hazard.

3:13:3 Solid Waste Pollution

Disposal of solid wastes has emerged as a very big challenge today. In developed countries, where availing human labour is highly expensive, advanced techniques and equipment are employed in recycling technology to tackle the problem of solid waste management. It is easier to recycle substances like paper cups, newspapers etc. but recycling heavy metals proves to be complex and problematic.

3:13:4 Controlling of Solid Waste Pollution

It is an urgent necessity to control solid waste pollution. Following measures are suggested in this regard.

i) Removal of Solid Wastes for Open-dumping

Solid wastes removal involves three steps: (a) Collection (b) Transportation and (c) Disposal. However in most settlements, dumping lines are absent. In many places where formal systems of garbage disposal are introduced, garbage is transported to open sites on the outskirts of the city and dumped to landfill sites. At these sites, the garbage decays and settles down. Though the city

becomes clean, its outskirts become dirty polluting the air, and ground water. Civic authorities often resort to this method, as it is the cheapest and involves expenditure only on transportation and transfer.

ii) Sanitary Land Filling

In this method, solid wastes are collected, transported and buried in deep trenches safely. In this method breeding of flies and mosquitoes are prevented; air is not polluted; and also there is no danger of catching fire. But during raining, the solid wastes dissolve in rain water and percolate into the ground polluting the ground water reserves. This could be avoided by spreading thick plastic sheets on the floor of burying trenches, then spreading the garbage and cover it with soil layers.

iii) Incineration

In this method solid wastes are put in the incineration plants and burnt. Incineration plants produce heat and electricity during incineration. After segregating the solid wastes, incineration should be attempted. As solid wastes are burnt in closed containers during incineration, no air pollution takes place. The slag getting deposited

at the bottom of petroleum, gum and plastic wastes too could be burnt in this method.

iv) Composting

Agricultural wastes could be subjected to composting and turn them into manures. There are three techniques involved in composting.

- a) Anaerobic composting is done in the absence of air. It takes a long time of 4 to 8 months.
- b) Anaerobic composting is done in the presence of air. It is completed in 2-3 weeks. It must be properly covered and a temperature of 55°C is to be maintained.
- c) Vermi-composting is done in the presence of air and earthworms. The obtained manure is colourless and rich in macro and micro nutrients. It can be dried and stored easily. For this method, controlled conditions of moisture and temperature are necessary.
- v) Organic wastes could be fermented to generate biogas that could be used for the production of electricity and as a fuel as well.
- vi) Decomposable solid wastes could be turned into substances that find alternate applications (e.g. waste clothes could be turned into door mats,

waste tyres could be used for providing bush in furnitures etc.) or recycled into other useful products.

3:13:5 Managing the Garbage Problem

Solid waste management involves the following techniques: (1) Reducing the generation of solid wastes (2) Repeated utilisation of substances (3) Recovery and recycling techniques (4) Finding alternate uses / applications for solid wastes (5) Pre-cycling.

1. Reducing the generation of Solid Wastes

- i) Reducing the use of packing materials like plastic and mettalic tapes, thermocoal board, cardboard boxes, paper wrappers etc.
- ii) Instead of paper cups and plates, metal tumblers and plates are to be preferred to avoid generation of paper wastes.
- iii) Reducing the use of plastic and polythene materials. However manufacturers and retailers prefer plastic over glass and porcelin because it is light weight, cheaper, micro wavable, durable and more resilient. Consumers like it because it is secure, strong and water-proof. Standard require that the products disposed of

must be non-toxic, derived from renewable resources, recyclable and biodegradable.

2. Repeated Utilisation

Substances like glass bottles, tincans, steel (eversilver) drums, jars, tumblers and utensils could be used repeatedly. Such materials should be preferred for our daily use.

3. Recovery & Recycling

- i) In Germany, highways are lined with sound barriers made out of old tyres, newspapers and construction debris.
- ii) Paper could be recycled to prepare paper boards, cardboard boxes, wall boards etc.
- iii) Among the metal wastes, useful metals like copper, iron and aluminium are segregated separately, after due refinement subjected to chemical processes and moulded into useful substances meant for day-to-day use.
- iv) Like metal wastes, plastic and rubber wastes could also be recycled. From plastics PVC pipes are manufactured; from rubber wastes hand gloves and matspreads could be prepared.

4. Finding Alternate Uses

In this method solid wastes could be directly put into alternate uses or byproducts. For example,

- i) ashes emitted by coal-fired power plants could be utilised in the preparation of cement and bricks.
- ii) from the crushed sugarcane left-outs, electricity and ethanol could be produced and the use of diesel may be minimised.
- iii) Using the old tyres, plastic materials and construction debris, sturdy roads that withstand water corrosion could be laid.

5. Pre-cycling

This is a new concept. It implies that we should consider the end-result of everything we buy and their impact on environment at the end.

3:13:6 Role of Individual in Preventing Environmental Pollution

As man is a 'highly civilised animal', he should respect the nature and learn to do away with things that are injurious to nature. Each and every individual has a part to play in the preservation and protection of the environment. We have the responsibility to

keep a green and clean environment. Individual responsibility towards the preservation and protection of environment are listed below :

- i) Preserving and protecting local water resources like tanks and lakes.
- ii) Avoiding the burning of plastic wastes (like during 'Bogi' festival) particularly in residential areas; protesting against establishing factories and industries in residential dwellings.
- iii) Finding alternate uses for solid wastes.
- iv) Restraining from throwing solid wastes in the drains.
- v) Handing over the collected garbage to the sanitary staff of municipality, after due segregation of biodegradable and non-degradables.
- vi) No dumping of construction wastes and animal wastes on the roadside; avoiding letting out sewage on the streets.
- vii) Burying the medical wastes safely.
- viii) In rural areas agricultural wastes could be composted to prepare manure and generate biogas to get electricity.

- ix) Helping for 'recovery and recycle' of solid wastes; impressing everyone to buy recycled goods.
- x) Avoiding as far as possible the use of polythene bags, synthetic fabrics, 'use and throw' items like ball pen, tea cups, sachet etc.
- xi) Reducing the use of fossil fuel as far as possible; instead of individual use, cars can be shared with friends and relatives for going to office or purchase; when it becomes necessary to travel alone, using public transport like bus and trains; taking a walk for short distances.
- xii) Cultivation could be attempted even in cities by establishing roof-top garden and using compost as manures.
- xiii) Implementing rain-harvesting in every building.
- xiv) Controlling the volume of sound emerging out from television sets and radios to be within safe limits.
- xv) Refraining from defecating the roadsides.
- xvi) Switching off the lights and fans when not required.
- xvii) Repairing the leaking taps, then and there.
- xviii) Recycling domestic water wastes and use it

for purposes other than cooking, drinking and bathing.

- xix) Resorting to organic farming by using biofertilisers and pesticides; avoiding ripening of fruits through the use of chemicals.
- xx) Fine-tuning the engines of vehicles to avoid fuel waste; avoiding noise pollution by fixing silencers and cleaning them periodically.
- xxi) Putting the garbage only in dustbins placed in street corners; avoiding littering the streets and other public places.
- xxii) Avoiding noise pollution due to the use of loud speakers during marriages, and other religious and social functions.
- xxiii) Avoiding the unnecessary wastes of food stuff and paper (using the double sides of paper for writing)
- xxiv) Growing plants in pots in balconies and terrace.
- xxv) Adopting small family norms to control population.
- xxvi) Developing public awareness about the need for and means of protecting environment from pollution.

3:14 Disaster Management

3:14:1 Meaning and Definition of Environmental Hazards

Environmental Hazards may be stated as those *extreme events* caused by *natural process* or man's activities which exceed the tolerable magnitude within or beyond certain time limits, make adjustment difficult, result in losses of property and lives.

The terms environmental *hazards*, environmental *stresses* and environmental *disasters* are used for extreme events such as earthquakes, floods, volcanic eruptions etc., which are caused by natural process or human activities.

The environmental *hazards* are the *abnormal processes* which cause environmental disasters. The intensity of environmental *disasters* are assessed in terms of damages done to the human society. The environmental quality is lowered and deteriorated by these *extreme events* which are termed hazards and *disasters*.

The environmental or ecological balance is disturbed; the resultant state of the highly disturbed natural environment is called '*Environmental Stress*'. The natural sudden physical processes and events

become hazards and disasters when humans live near a danger.

3:14:2 Types of Environmental Hazards and Disasters

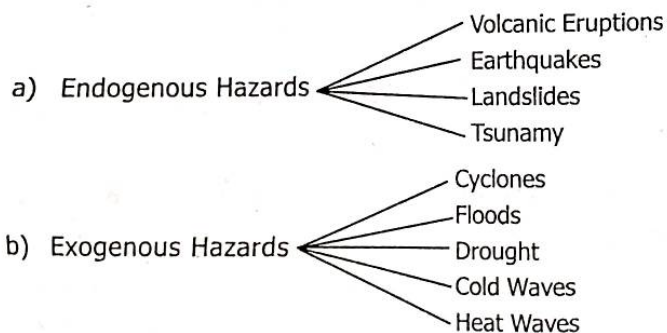
The extreme events are divided into two categories, on the basis of causative factors.

- (i) Natural Hazards
- (ii) Man-induced Hazards

A brief description of important natural hazards / disasters and their management have been given in the sections to follow.

3:14:3 Natural Hazards and Disasters

These involve rare high-intensity processes and extreme events caused by 'terrestrial or endogenous' hazards and 'atmospheric or exogenous hazards'. Thus natural hazards are of two types. They are :



3:14:3:01 Earthquake

(A) Occurance of Earthquake

An earthquake is the violent shaking of the Earth's surface caused by rapid movement of the Earth's rocky outer layer. Deep in the Earth are rocky layers which are constantly moving. As these rocky layers push one another, causing collisions at times during which their edges crack and the enormous pressure developed during collision escapes through the crack developed. The escaping pressure through the crack move out and make the upper layers of earth tremble. These trembling / vibrations among the rocky layers, are not felt by us, if they are small. But if the collision between the rocky layers happens to be very powerful, rocky layers break and get destroyed and the resultant massive pressure developed act on all sides in a bid to escape which causes massive shaking of Earth's upper surface called earthquake.

The place of origin of an earthquake is called '**focus**'. The point on the Earth's surface vertically above the focus is called the '**epicentre**'. The intensity of vibration is maximum near the epicentre.

The resultant energy waves spreading on all sides from the focus are called '**seismic waves**'.

(B) Causes of Earthquake

- i) Earthquakes are caused by tectonic forces.
- ii) Earthquakes are generally due to the sudden movement of the rocks strata, along faults or fractures in the crust.
- iii) Earthquakes are also caused by volcanic eruptions.
- iv) Sometimes water percolates so deep into the Earth that it turn into steam. This steam tries to force its way out which causes earth tremours.
- v) Earth tremours may also be produced by man by disturbing the balance in the earth's crest. Eg. dam construction (e.g: earth quake at Koynanagar Township) and underground nuclear explosions.

(C) Adverse Effects of Earthquake

- i) Many villages and towns are destroyed and there is considerable loss of life and property.
- ii) Some places are submerged under the sea.
- iii) Big cracks and fissures are developed on the earth surface which disrupt communication.

- iv) Rivers disappear and change their course.
- v) Railway lines are twisted.
- vi) Houses and buildings develop cracks and fissures during mild tremours; massive earthquake destroys the dwellings of all types.

(D) Beneficial Effects of Earthquake

1. New springs are formed which help irrigation.
2. Submerged coastal land emerge out of water and makes fertile plain.
3. Precious metal and mineral come up to the surface.
4. When there is eruption due to earthquake, it causes lava to flow out, which enrich the fertility of the soil.
5. Many lakes, plateaus and islands are formed due to earthquakes.
6. Violent earth quakes have led to the formation of hills and mountains which help in causing rainfall.

(E) Some Symptoms of Earthquake

1. Seawater recedes from the seashore; water level in wells may either significantly fall or rise.

2. Cracks will be developed in the surface of the earth near the earthquake region.
3. Radio active radon gas may emerge from wells.
4. Level of hydrogen gas on land may increase.
5. Sudden flashes of lightening with thunderous sound may appear in the sky.
6. Windows and doors of buildings may vibrate violently; utensils from cup boards may fall and roll; cracks may appear in walls; we may feel on the floor under our feet moving; in some places springs may be developed.

(F) Precautionary Measures

Earthquakes can neither be prevented nor controlled; but could be predicted and suitable precautions followed to safeguard life and property. Some of the precautionary measures are :

- i) Stay in open spaces where there are no buildings or trees in the nearby.
- ii) Inside the house, take shelter under the tables.
- iii) Standing under the lintel above the doors saves us to some extent from building debris, if the building crashes.
- iv) Fire is not to be used during earth quakes.

Earthquakes are measured in 'Rictor Scale'. Earthquakes above 6 Rictors may cause damage to life and property. A massive earthquake struck at 'Bhuj' in Gujarat on 26.01.2001, killing atleast 17,000 people and more than 7,00,000 rendered homeless. Numerous towns and villages were reduced to rubble including Bhuj. Since 1900, about 20 massive earth quakes have taken place all over the world.

(G) Rehabilitation Measures to be Undertaken in Earthquake - Hit Regions

Rehabilitation measures to be undertaken by the Government and local authorities with the co-operation of the public are : **(i) Searching (ii) Rescuing and Saving the Victims.**

- 1) Those who have been rescued should be offered first aid immediately and then sent to medical camp to provide necessary medical care
- 2) Arranging provision for medical and food facilities
- 3) Dead bodies are to be retrieved and reached out to the Kith and Kin of the dead
- 4) During search, accustic devices could be used to detect the groaning sound of those buried alive under the debris. Infra-red Cameras

could also be used during search for people or property.

Relief teams should try to receive constantly reports and messages from the government, media and the public and act appropriately according to the circumstances.

State Governments should enlist the active help of the Central Government and receive the assistance of '**National Disaster Rehabilitation Centre**' in saving people from natural disasters as well as notifying warnings about the impending dangers well in advance so that relief measures could be arranged for. Satellite phones could also be used for effective communication during the period of natural disasters when other modes of communication become disrupted.

3:14:3:02 Tsunami

(A) Occurance of Tsunami

The term 'Tsunami' originated from Japan, which is prone to frequent earthquakes. It is derived from the Japanese word 'Tsu' meaning port and 'nami' meaning wave-attack. Tsunami refers to gigantic waves rising to a height of 20 to 30' attacking coastal

regions of even far off places and inflicting massive destruction, are mainly caused by severe earthquake or volcanic eruptions. Tsunami waves originating from the 'focus' of earthquake travel at a speed of 700 to 800 K.M. per hour and reach the coasts in few minutes. Following the attack of the first tsunami wave, a series of waves will be going on coming and gradually their intensity and frequency gets reduced. On 26.12.2004, at 6.45 a.m. a severe earth quake measuring 8.9 in the rictor scale, occurred with its epicentre located in Sumatra Island of Indonesia and as a result of this tsunami waves were generated which reached Tamilnadu Coast at about 8.45 A.M. inflicting heavy loss of life and property.

(B) Tsunami Information Centre

In 1965, with the efforts of UNESCO, a '**Tsunami Information Centre**' was established at **Honalulu** in the **Howaii** Island to get in constant touch with Science Research Centres, Institutions of Higher Education, Military Head Quarters etc. share Tsunami related data with member countries, and provide advance notifications and warnings. In the library of the Tsunami Information Centre, relevant data that are collected from across the world are deposited and preserved.

(C) Measures to Manage Tsunami

Before the Occurance of Tsunami

- i) Knowing whether one's apartment, office or educational institution is located in the tsunami - prone region.
- ii) Avoiding buildings and apartments within a distance of 150 ft from the sea shore.
- iii) Knowing in advance, the safe quick routes for exit to escape from tsunami, if at all it occurs.
- iv) Preparing the list of essential personal belongings that are to be carried away safely in the event of any tsunami attack.

During Tsunami

- i) On coming to know about the impending tsunami, one should inform about it to all his family members.
- ii)* On getting the warning news broadcast by television and radio and messages through several media, people in the tsunami - prone regions should move away quickly to places of safety, carrying the essential personal belongings.
- iii) People living in coastal regions, if happen to feel massive land vibrations should try to move

to places located in the high regions in that area. They should not wait for the notification from the Government.

After the Tsunami Attack

- i) Trying to know about the events that are taking place after the attack of tsunami through mass media like television and radio.
- ii) Coming forward voluntarily to safeguard children, the handicapped and old people by helping them to move out to places of safety.
- iii) Visiting the places of disaster should be avoided as it may cause bacterial attack.
- iv) Getting the required medical care.
- v) Using cell phones for emergency needs only; during disaster power distribution breaks down, rendering mobile-phone charging not possible.
- vi) Visiting places inundated by tsunami water is dangerous; due to continued stagnation of water, chances are there for building bases to get eroded and collapse at any time.
- vii) Co-operating with the government and voluntary agencies in their rescue and rehabilitation work.

3:14:3:03 Landslides

A landslide is the descent of a mass of earth and rock down a mountain slope, under the influence of gravitational force. Landslides are much more faster than flows and sometimes roar down slope with a thunderous sound and great force. Water plays a lesser role, although it does help in the weathering and loosening that has gone before. A landslide is in effect a collapse and no lubricants are necessary, although the weight of water saturation certainly does contribute. Landslide is a common feature during rainy season on the Mettupalayam - Ooty Ghat Road and the Himalayan Roads from Rudhraprayag to Bhadrinath and Kedarnath.

(A) Causes of Landslides

- i) Earth tremors and incessant rains are the major factors of landslide.
- ii) Buildings constructed on the upper reaches of slopes cause heavy pressure underneath due to which landslide may occur.
- iii) Due to soil erosion when base layers become thin, vacuum is created, towards which bedrock as well as over burden rush causing landslide.
- iv) Deforestation, tunneling and plaining sloppy

rocks for laying railtracks and roads, building apartments, undertaking cultivation in hillslopes etc. also contribute significantly to landslides.

(B) Preventing Human Activities Causing Landslides

- i) Denying permission for construction of building on rock and mountain slopes.
- ii) Constructing a strong concrete blocking wall at the bottom of the slope.
- iii) Strengthening bedrocks with gravel.
- iv) Forbidding cultivation of crops on hill and mountain slopes.
- v) Removing the vegetations along the roadsides in hilly regions and forbidding destruction of forest cover to develop cultivation.

However, landslides due to earth tremours can not be controlled.

3:14:3:04 Volcano Explosion

In hot places, under a depth of 120 to 160 K.M. below the surface of earth, molten rocks in the form of liquid, called '**Magma**' with hot gas exist. Due to pressure changes when it rises up and is at a depth

of 40 K.M. from earth's surface, greater pressure surrounding 'Magma' further pushes it up, during which it melts rocks on its way and get stabilised at a depth of 3 K.M. depth from earth's surface in the form of a conical chamber, from which it ejects out with great force through weak spots on the surface of earth as an explosion with thunderous noise. The redhot liquid '**lava**' containing molten rocks and solid stones resembling smouldering coal pieces along with gaseous clouds rise up to the sky, presenting a show similar to that of fireworks. The spot from where 'lava' ejects out is called '**Central Vent**'. Burning stone and other substances emitted through the central vent, get cooled and heaped like a mountain around it. The top of the mountain so formed will have a bow like carter. This mountain with carter is known as '**Volcanic Mountain**'. Eruptions get cooled and the volcanic mountain remains dormant for some years, after which again it starts smoking and erupting, the temperature of which will be not less than 1100°C . Volcanic mountains are present in Japan, Indonesia, Phillipines, China and Greece.

(A) Destructive Effect of Volcanoes

- i) Lava flowing down a volcano destroys the human settlement and render the land useless.

- ii) Ashes and dust puffed out by volcano may bury villages and towns near it; cause suffocation and breathing troubles in people.
- iii) Disrupts air traffic.
- iv) Inflicts heavy loss of life and property.

3:14:3:05 Floods

Floods are rising level of water along a river channel or coast leading to inundation of land that is not normally submerged.

Flood results from an unfavourable combination of meteorological conditions and the physical conditions of the drainage basin. Floods in coastal areas are due to the rise in the level of sea water due to the following reasons :

- i) **Unusual meteorological conditions :** During a cyclone, sea becomes rough and water level increases. Sheets of sea water may breach the seashore and enter into the coastal land when the cyclone crosses the seashore.
- ii) Due to volcanic explosion and earthquake under the sea, giant tsunami waves are generated which on reaching the coasts floods the land.

The causes for floods due to swollen rivers are

many of which the following are the important ones.

- i) Continuous heavy rainfall during the monsoons and cloud bursts.
- ii) Excessive melting of snow and ice at the upper reaches of mountains where the river originates.
- iii) Bursting of dams built across the rivers.
- iv) Erosion of river banks leads to its breaching and river water enters into village / town.
- v) Landslides near the river banks too causes flood.

(A) Effects of Flood

Floods affect the normal life of people and livestock; Crops, buildings and roads are damaged; communication gets disrupted; capacity of rivers gets decreased due to excessive siltation, raising the river bed.

(B) Measures to Control Flood

- i) Strengthening the river banks and raising their height.
- ii) Soil protection is ensured in catchment areas of rivers.
- iii) Constructing more number of dams and check dams across the rivers.

- iv) Evacuation of people from flood-prone areas and settling them in safer places.

3:14:3:06 Cyclone

A) Meaning of Cyclone

Cyclone is a region of low atmospheric pressure surrounded by a wind system, in a counter clockwise direction. The term has often been loosely applied to storm and disturbance attending such pressure systems, particularly the violent tropical hurricanes and tyoons, which centre on areas of unusually low pressure. As storms in desert rains sand, they are called '**sand-storms**'; similarly in high altitude mountain regions, due to severe atmospheric pressure changes, storms rain snow and ice. They are called 'snow-storms' or '**blizzards**'.

Generally by cyclone we refer to, severe atmospheric pressure changes generating low pressure zone in the sea as a result of which strong wind blows towards the land, accompanied by copious rainfall. During cyclonic storms, wind blows strongly with a speed of 100 to 180 K.M. per hour. There will be heavy rainfall when the cyclone approaches the sea-shore. After crossing the seashore its speed gradually decreases and finally dissipates.

Atmospheric pressure changes are responsible for the generation of cyclone and its intensity and speed depend on the magnitude of pressure changes that took place.

B) Damages Caused by Cyclone

Due to cyclonic storms, villages and towns located on the coastal areas are flooded with water; trees are uprooted; crops, buildings, roads, communication structures and equipment get damaged. Sometimes land slides too occur. Thus cyclonic storm greatly affects the environment.

Meteorological centres detect the appearance of a cyclone in the sea (by assessing the cloud cover, changes in atmospheric pressure, direction and intensity of wind etc. with the help of radar and satellite pictures received) judge the location of its eye, its direction of movement and speed. Using these data they predict when and where the cyclone will cross the sea coast and notify advance warnings to the public, enabling people to take necessary precautions, so as to avoid loss of life and property. Government should make necessary arrangements for people to stay in safer places.

India has been stormed with 50 severe cyclones

during the period 1900 to 1999. Each severe cyclonic storm inflicts an economic loss of 200 to 300 crore rupees. Normalcy does not return till the rehabilitation work gets completed as a result of which other developmental projects are affected. Cyclones are mostly formed in the Bay of Bengal and it is in the ratio 4:1 as compared to Arabic sea. As sandstorms and snowstorms occur well outside the regions of human habitats, they do not cause much damage. Snow-storms may affect the mountaineers. But cyclone storms occur near thickly populated areas and hence cause heavy loss of life and property.

C) Measures to Control Cyclone Damages

1. Intensity of the speed of wind during cyclones get reduced if there are blocks in its way in the form of hills or tree covers and hence the resultant damage will be less. Therefore it is advised to grow vegetative bushes and tree covers on the sea coasts and estuaries.
2. It is better to construct buildings on the plains in the hilly regions than on high rise areas near the top (where the speed of wind is usually high).
3. Impact of cyclone on buildings constructed in a circular or polygon shape will be less.

4. Instead of fixing big windows and doors in the building, it is preferable to have windows and doors fitted with small glass plates.
5. Advance notification of warning about the impending cyclone helps the public take precautionary measures to safeguard the life and properties.
6. Power supply could be stopped while cyclone crosses the seashore; road and air traffic could also be suspended.
7. **'Cyclone Rehabilitation Centres'** could be established on the sea coasts so that those who are likely to be affected could be made to stay during cyclones.

3:14:3:07 Oil Spills

An oil spill is the release of a liquid petroleum hydrocarbon into the environment, especially the marine ecosystem, due to human activity and is a form of pollution. The term is usually given to marine oil spills, where oil is released into the ocean or coastal waters, but spills may also occur on land. Oil spills may be due to leakages of crude oil from tankers or pipeline break, big oil tanker ships sink due to accident or enemy attack, or drilling operations go wrong in offshore platforms.

Generally oil spills harm ocean life in two ways viz. **Fouling or Oiling** and **Oil Toxicity**.

Fouling or Oiling occurs when oil physically harms a plant or animal. Oil can coat a bird's wings and leave it unable to fly or strip away the insulating properties of a sea mammal's fur, putting it at risk of hypothermia. The degree of oiling often impacts the animals' chances of survival.

Oil consists of many different toxic compounds. These toxic compounds can cause severe health problems like heart damage, stunted growth, immune system effects and even death to sea and ocean living plants, birds and animals.

The trained experts use different equipment and tactics to contain or remove oil spill from the environment. Booms are floating physical barriers to oil, which help keep the oil spill contained and away from sensitive areas like beaches, mangroves and wetlands. Skimmers are used to "skim" oil from the sea surface. In-situ burning or setting fire to an oil slick can burn the oil away at sea and chemical dispersants can break up oil slicks from the surface. clean up activities can never remove 100% of the oil spilled.

Oil spills can have disastrous consequences for society - economically, environmentally, and socially. As a result, oil spill accidents have initiated intense media attention and political uproar, bringing many together in a political struggle concerning government response to oil spills and what actions can best prevent them from happening.

Conclusion

Concepts of 'Environmental Degradation', 'Environmental Hazards', Differences between these two concepts, Types of Environmental Degradation, Meaning and Definition of Environmental Pollution, Different Types of Environmental Pollutants, Basis of the cost of Environmental Pollution, Important types of Environmental Pollution viz. land, air, water, noise, light, radiation and solid waste, their sources, measures to control them etc. have been discussed in detail. Further important disasters like earthquake, tsunami, landslide, volcano explosion, flood and cyclone and oil spills and their management, have been explained in detail.

Review Questions :

1. What do you mean by 'environmental degradation'? (C) [Ans: 3:01:1]

2. Differentiate 'environmental pollution' from 'environmental degradation' (C) [Ans: 3:01:2]
3. What are the causes of environmental degradation? (C) [Ans: 3:01:3]
4. Mention the different types of environmental degradation (C) [Ans: 3:01:4]
5. Define the term 'Pollution' (C) [Ans: 3:02:1]
6. Explain the different kinds of environmental pollutants (B) [Ans: 3:02:2 + 3:02:2:01 + 3:02:2:02]
7. What do you mean by 'Soil Pollution'? (C) [Ans: 3:05:1]
8. State the sources of soil pollution. (C) [Ans: 3:05:2]
9. Mention the different types of soil pollution. (C) [Ans: 3:05:3]
10. Mention the effects of soil pollution. (C) [Ans: 3:05:4]
11. Explain the measures to control soil pollution. (B) [Ans: 3:05:5]
12. What is soil pollution? What are its effects? Discuss the measures to control soil pollution? (B) [Ans: 3:05:1 + 3:05:2 + 3:05:4 + 3:05:5]
13. What do you mean by 'water pollution'? (C) [Ans: 3:06:1]
14. State the reasons for water pollution. (C) [Ans: 3:06:2]
15. Mention the different kinds of water pollution. (C) [Ans: 3:06:3]
16. Discuss the different types of 'water pollution' (A) [Ans: 3:06:1 + 3:06:3 + 3:06:3:01 to the end of 3:06:3:05]

17. What are the adverse effects of water pollution? (C) [Ans. 3:06:4]
18. Explain briefly the ways and means of controlling water pollution (B). [Ans: 3:06:5]
19. What do you mean by 'Air pollution' (B) [Ans: 3:07:1]
20. Explain the sources of air pollution? (B) [Ans: 3:07:2]
21. Mention the different types of air pollution. (B) [Ans: 3:07:3]
22. Discuss the different types of air pollution. (A) [Ans: 3:07:3 + 3:07:3:01 to the end of 3:07:3:04]
23. State the adverse effects of air pollution. (C) [Ans. 3:07:4]
24. Discuss the ways and means of controlling air pollution. (B) [Ans: 3:07:5]
25. What is air pollution? Discuss the different types of air pollution and the measures to control them. (A) [Ans. 3:07:1 + 3:07:3 + 3:07:3:01 to the end of 3:07:3:04 + 3:07:5]
26. What is meant by 'marine pollution'? (C) [Ans. 3:08]
27. How does marine pollution takes place? (B) [Ans. 3:08:1]
28. How does the sea gets polluted? Explain the adverse effects of marine pollution and the measures to control marine pollution. (A) [As. 3:08 + 3:08:1 + 3:08:2 + 3:08:3]
29. Write a short note on each of the following :
 - i) Measures to control marine pollution (C) [Ans. 3:08:2]

- ii) Noise Pollution (C) [Ans. 3:09:1]
- iii) Sources of marine pollution. (C) [Ans. 3:08:1]
- iv) Thermal Pollution (C) [Ans. 3:10]
30. Mention the sources of noise pollution. (C) [Ans: 3:09:2]
31. What are the effects of noise pollution on human beings? (C) [Ans: 3:09:4]
32. Explain the measures to control noise pollution. (B) [Ans. 3:09:5]
33. How could thermal pollution be controlled? (C) [Ans. 3:10:2]
34. What is radiation pollution? Mention its sources and effects. How could radiation pollution be controlled? (B) [Ans: 3:11 + 3:11:1 + 3:11:2 + 3:11:3]
35. Write a short note on each of the following:
 - i) Adverse effects of thermal pollution. (C) [Ans. 3:10:1]
 - ii) Light pollution. (C) [Ans. 3:12:1]
 - iii) Effects of light pollution. (C) [Ans. 3:12:3]
 - vi) Oil Spills (C) [Ans: 3:14:3:07]
36. Explain the sources and types of light pollution. (B) [Ans: 3:12:2 + 3:12:4 + 3:12:4:01 to the end of 3:12:4:05]
37. What is solid waste pollution? Mention its sources. (C) [Ans: 3:13 + 3:13:1]